

COMS 4720 HW6

1) Let us make some column equations where each column adds digits from right to left.

Coloumn	Equation
Ones	$O + O = R + 10 \cdot C_1$ (which gives a carry over of c_1)
tens	$W + W + C_1 = U + 10 \cdot C_2$ (which gives a carry over of C_2)
Hundreds	$T + T + C_2 = O + 10 \cdot C_3$ (which gives us a carryover of C_3).
Thousands	C_3

We can observe in the table that when we added 2 3-digit numbers, it gave us an output of a 4-digit number, which carries over C_3 to thousands and gives us the result F . Since C_3 is 1 in this case, we can say that F will be 1 as well.

Now, let us plugin this value in the hundreds equation, so we get $2T + C_2 = O + 10 \cdot C_3 = O + 10$, which leaves us with $O = 2T + C_2 - 10$. Now, let us test some possible values for T and C_2 , such that O is a single digit.

Value of T	Value of C_2	Value of O
5	0	0
5	1	1
6	0	2
6	1	3
7	0	4
7	1	5
8	0	6
8	1	7
9	0	8

9	1	9
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Now, since F and O must be distinct, so we can say that O is not equal to 1.

Now, let us work on the ones column next, so that we can compute the values for R and C1.

O	2O	C1	$R = 2O - 10 \cdot C1$
2	4	0	4
3	6	0	6
4	8	0	8
5	10	1	0
6	12	1	2
7	14	1	4
8	16	1	6
9	18	1	8

Now, by looking at the table we can say that we have a safe bet if we assume that $T=7$, $O=4$ and $C2 = 0$, since all we need is that these numbers should be distinct from F which is 1, so we have got all unique numbers till now, that are $\{1, 7, 4, 8\}$

Now, let us look at the tens column and try and extract some information for this question. Over here we know that $C1$ and $C2$ are equal to 0 and $2W = U$, which means that U is an even integer, so which means that we can get the following pairs:

W	U
3	6
4	8
2	4

Since, we have already used $O = 4$ and $R = 8$, we can say that $W = 3$ and $U = 6$ in this case. Now, let us try and verify the result that we just got, that is, c.

We already know that $734 + 734 = 1468$, where $F = 1$, $O = 4$, $U = 6$ and $R = 8$ which verifies that our answer is correct.

Hence, the final answer for this question is $T=7$, $W=3$, $O=4$, $F=1$, $U=6$, $R=8$.

2a) Since we have 5 houses and each one of those are distinguished on the basis of 5 categories, we can say that there are going to be 25 variables in total, which can be defined as follows:

Color: Color_Yellow, Color_Blue, Color_Ivory, Color_Green, Color_Red

Nationality: Nationality_Norwegian, Nationality_English, Nationality_Ukrainian, Nationality_Japanese, Nationality_Spainard

Drink: Drink_Water, Drink_Tea, Drink_Milk, Drink_Coffee, Drink_OrangeJuice

Candy: Candy_KitKat, Candy_Snicker, Candy_Smarties, Candy_Hershey, Candy_MilkyWay

Pet: Pet_Horse, Pet_Dog, Pet_Snails, Pet_Zebra, Pet_Fox

Houses: 1,2,3,4,5 (Given in the question)

b) 1) Nationality_English = Color_Red

2) Nationality_Spainard = Pet_Dog

3) Nationality_Norwegian = 1

4) Color_Green = Color_Ivory + 1

5) Candy_Hershey = Pet_Fox + 1

6) Candy_KitKat = Color_Yellow

7) Color_Blue = Nationality_Norwegian + 1

8) Candy_Smarties = Pet_Snails

9) Candy_Snickers = Drink_OrangeJuice

10) Nationality_Ukrainian = Drink_Tea

11) Nationality_Japanese = Candy_MilkyWay

12) $\text{Pet_Horse} = \text{Candy_KitKat} + 1$

13) $\text{Drink_Coffee} = \text{Color_Green}$

14) $\text{Drink_Milk} = 3$

3) Since we are talking about the CSP setup, we know that we are going to have variables such as WA, NT, SA, Q, NSW, V, T and domain which has red, green and blue colors. Moreover we have a constraint which says that the regions must differ in color. Also, from the question we know that $\text{WA} = \text{green}$ and $\text{V} = \text{red}$.

Now, let us start doing the problem.

We know that from $\text{WA} = \text{green}$ and $\text{V} = \text{red}$, we can say that we can enforce consistency from SA by saying that SA is not same as WA (as they have different colors) so we can remove green from the domain and similarly we can remove red from the domain since V and SA are not same as well, which only leaves us with blue in the domain.

Now, in our next step we are going to remove NT, Q and NSW's domain which leaves us with the following:

$\text{Dom}(\text{NT})$ (Domain of NT) = $\text{Dom}(\text{NSW}) = \text{Dom}(\text{Q}) = \{\text{red}, \text{green}\}$

Next, we will enforce consistency on NT with neighbor WA which is green by saying that NT is not green anymore, which leaves NT with only red in its domain.

Next, we will propagate NT to Q by saying that Q does not have red anymore in its domain which means it is only left with green in its domain.

Next, we will enforce consistency on NSW from all of its neighbors, by saying that NSW and SA don't have same colors, NSW and V don't have same colors and NSW and Q don't have same colors, which means that there are no colors left in NSW's domain.

Hence, since the AC-3 empties NSW's domain, we can say that the partial assignment is inconsistent under arc consistency.